Magnetism of thin Ising films with rough surfaces

Aastha Arya Sonam Kumari

Abstract

This study examines ferromagnetic Ising thin films whose thickness varies randomly, using Monte Carlo simulations to investigate their magnetic behavior. A simple cubic lattice with lateral dimensions N is used to represent the films. One surface remains flat, while the local thickness is assigned from a discretized Gaussian distribution characterized by a mean L and an rms deviation ΔL. The roughness pattern considered here is: the types of films in which ΔL is constant for all thicknesses L. It is observed that increasing the surface roughness leads to a decrease in the critical temperature and a suppression of the specific-heat peak, while the susceptibility peak remains unchanged for small roughness ($\Delta L \leq 1$) and declines only for larger values of ΔL. The resulting shift in the critical temperature, $t_L = T_C(3D) - T_C(L, \Delta L)$, is studied for films with different roughness.

1 Introduction

Magnetism has been explored for centuries, starting with William Gilbert's De Magnete (1600), which laid the groundwork for future studies. Recent research has focused on nanoscale magnetic systems, particularly films with a thickness of a few nanometers. These ultrathin structures—often consisting of a few atomic layers—demonstrate magnetic properties that significantly differ from those of bulk materials owing to reduced dimensionality and interface effects. Their magnetic properties are considerably affected by their structural features, which are in turn influenced by growth processes that naturally introduce features like thickness variations.

We study a model of Ising spins with varying thicknesses to investigate the effect of surface roughness on the magnetic properties of thin films. We assume that changes in magnetic behavior are primarily due to uncorrelated surface roughness in the model.

We use the Ising model to represent the magnetic interactions in the system. In this framework, the magnetic lattice consists of N fixed sites arranged in an n-dimensional periodic lattice ($n = 1, 2,$ or 3), with each site carrying a spin variable s_i ($i = 1, \ldots, N$) that can take the value $+1$ (spin up) or -1 (spin down). These spin variables are the only degrees of freedom, and the complete state of the system is fully specified by the set $\{s_i\}$, which describes the spin configuration of the entire lattice.

This work is organized as follows. In Section 2, we describe the simulation and calculations performed, including the model used and the Monte Carlo method. Section 3 presents the Ising model, where we discuss Onsager's result. In Section 4, we present the results and discussion, including plots,

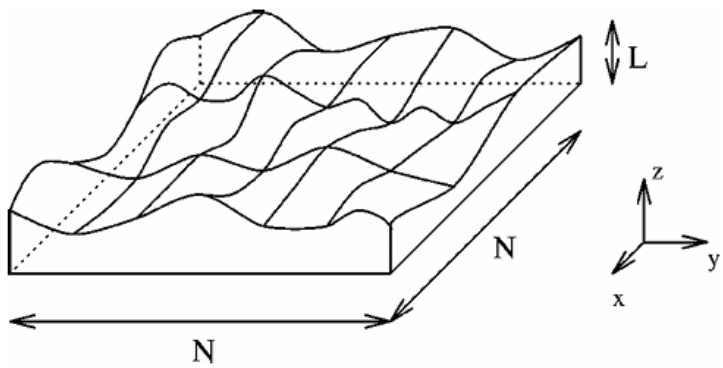

Figure 1: Thin film with length and width N, one flat surface and one rough surface with mean thickness L.

the thermodynamic behavior of the films, the effects of surface roughness on physical quantities, and how the critical temperature T_c varies with increasing roughness. Finally, Section 5 summarizes our main conclusions.

2 Simulation & Calculation

2.1 Model

We considered thin film of lateral size $N \times N \times L$ where L has spatially thickness $h(x, y)$. It has simple cubic lattice structure, where spin sits for size $L \leq h(x, y)$. We generated random rough height profile $h(x, y)$ with Gaussian fluctuation, with mean thickness in z direction and for roughness we took absolute RMS deviation. Also the height is uncorrelated and to avoid negative or large value we set $0 \leq h(x, y) \leq 2L_{\text{mean}}$. We built a 3D spin array of shape (N, N, H_{max}) where $s_{i,j,k} \in \{+1, -1, 0\}$.

$$H = -J \sum_{\langle i,j \rangle} S_i S_j - H \sum_i S_i \tag{1}$$

where $\langle i, j \rangle$ denotes a pair of nearest neighbors, $S_i = \pm 1$ is the spin variable at site i, and H is an external magnetic field. We considered nearest neighbor spin interaction, ferromagnetic coupling J. Here we have considered zero magnetic field, free boundary condition in z and periodic boundary condition in lateral direction (x, y). We studied films with $N = 15, 25, 35, 40, 45$ with $L = 2$ and $\Delta L = 0, 2/3, 1$. We performed Monte Carlo simulation using Metropolis algorithm, where we considered zero magnetic field. For equilibration 5×10^4 initial Monte Carlo steps used at each temperature. We need to take average of thermodynamic quantities so we took 10^4 spin configuration. Since this experiment is for uncorrelated roughness we skip 20-30 Monte Carlo steps between two successive configuration. One needs different realization for averaging, instead we did

for single realization as we introduced disorderness and periodic boundary condition to take care of averaging.

2.2 Calculation

In Metropolis algorithm we flip the spin and see if it reduces the energy, flip is accepted whereas if it increases the energy it accepts with probability $e^{-\Delta E/T}$. As each spin wants to align with its neighbors and flipping it breaks some bonds and create other bonds. This change in energy gives the cost.

$$\Delta E = 2Js\sum_i s_i + 2\mu Bs \tag{2}$$

where i is sum over neighbors which include 4 in-plane neighbors and 2 vertical neighbors. And B is the magnetic field. We also need total number of active sites that is number of spins for normalizing thermodynamic quantities $N_{\text{active}} = \sum_{x,y} h(x,y)$.

Energy is computed as

$$H = -J\sum_{i,j} S_i S_j \tag{3}$$

Magnetization is computed as

$$m = \frac{1}{N_{\text{active}}}\sum_{i\,\text{active}} S_i \tag{4}$$

Then specific heat is computed as

$$C_v = \frac{1}{N_{active}}\frac{\langle E^2\rangle - \langle E\rangle^2}{k_B T^2} \tag{5}$$

Then Susceptibility is computed as

$$\chi = \frac{1}{N_{active}}\frac{\langle M^2\rangle - \langle M\rangle^2}{k_B T} \tag{6}$$

3 2D Ising Model

3.1 Onsager's Result

The 2D Ising model is exactly solvable in absence of external magnetic field. We are working on 3D model, where x and y direction is fixed and z direction is varying. So for 2D we need $L = 0$ in z direction.

$$H = -J\sum_{i,j} S_i S_j \tag{7}$$

Since magnetic field is zero, system should be in paramagnetic phase. At low temperature $\beta J \gg 1$. Since square lattice has 4 nearest neighbors, so there are in total $2N$ pairs of nearest neighbors. If all spins are aligned in same direction, each pair will contribute energy $-J$, so energy per spin will

be $-2J$. According to Onsager result, the low temperature asymptote $U/N = -2J$ signifies all spins are aligned at low temperature which means system behaves as ferromagnet.

Onsager's Solution is given as:

$$(\sinh 2\beta J - 1)^2 + (2 - \cos\theta_1 - \cos\theta_2)\sinh 2\beta J \tag{8}$$

At $\beta = \beta_c$, $\sinh 2\beta_c J = 1 \implies k_B T_c = 2.269J$

4 Results and Discussion

4.1 Plots

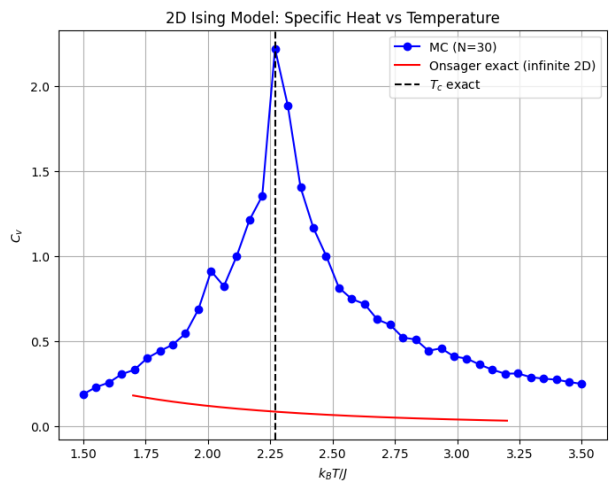

Figure 2: Onsager Result

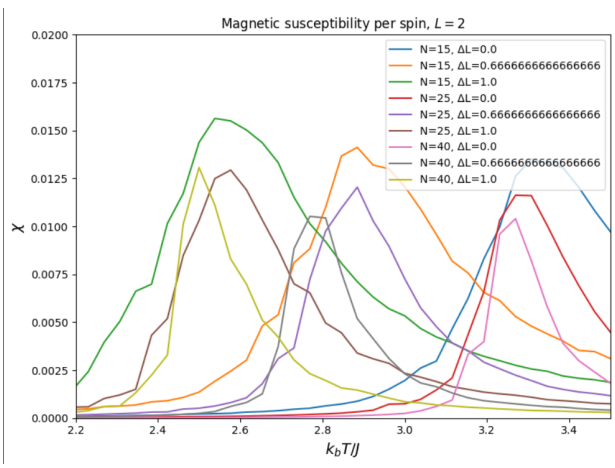

Figure 3: Magnetic Susceptibility

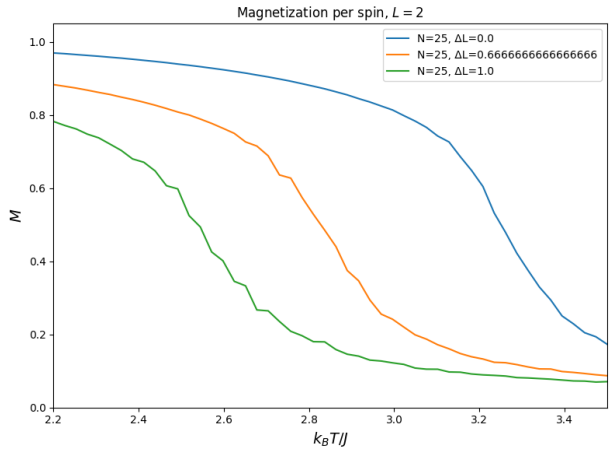

Figure 4: Magnetization

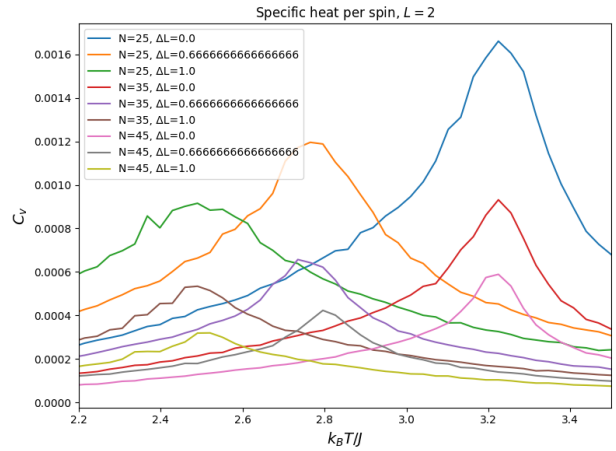

Figure 5: Specific Heat Capacity

4.2 Thermodynamic Behavior

To study how surface disorder affects magnetic behavior, we examined the susceptibility, specific heat, and magnetization of Ising films with different levels of roughness. A phase transition occurs when there is a singularity in the free energy or one of its derivatives. Specific heat and magnetic susceptibility correspond to second order derivatives of the free energy (with respect to temperature and magnetic field, respectively) and diverge at the critical point in an infinite system, while magnetization, being a first derivative with respect to the magnetic field, changes continuously. In our simulations of finite systems, the divergences in susceptibility and specific heat are smoothed into rounded peaks. Therefore, even though the true critical temperature is defined only in the thermodynamic limit ($N \to \infty$), in practice it is estimated from the peaks observed in the largest simulated systems.

Figure 3 shows the temperature dependence of the magnetic susceptibility for films with varying surface roughness. As roughness increases, the susceptibility peak shifts to lower temperatures, indicating a decrease in the critical temperature $T_c(L, \Delta L)$. For small roughness ($\Delta L \leq 1$), the susceptibility peak remains largely unchanged, indicating that moderate disorder has only a limited effect on magnetic correlations. Significant reductions in susceptibility occur only at larger roughness values, demonstrating that strong disorder increasingly reduces correlations across the film.

A similar trend is observed in the specific heat, shown in Figure 4. The temperature of the specific-heat peak also move towards lower values with increasing roughness. While the peak height decreases, the total area under the curve remains approximately constant, suggesting that thermal fluctuations are redistributed rather than lost.

Magnetization per spin, shown in Figure 5, changes gradually and continuously, reaching a maximum at low temperatures and vanishing above the critical temperature. As roughness increases, the magnetization decreases, following a similar pattern.

4.3 Effect of Film Roughness on Critical Temperature

The impact of surface roughness on films with a constant mean thickness L can be measured by the relative reduction of T_c:

$$\epsilon(L, \Delta L) = \frac{T_c(L, \Delta L) - T_c(L, 0)}{T_c(L, 0)}$$

where $T_c(L, \Delta L)$ is the critical temperature of the rough film and $T_c(L, 0)$ is the critical temperature of the smooth film. As the mean film thickness L increases while the roughness ΔL is kept constant, the relative decrease in the critical temperature becomes smaller. The reduced critical temperature t_L is expected to scale as

$$t_L = \frac{T_c(3D) - T_c(L, \Delta L)}{T_c(3D)} = AL^{-\lambda}$$

where $\lambda = 1/\nu$ and ν is the three-dimensional critical exponent of the Ising universality class.

However, for small values of L, this three-dimensional scaling is not observed, even in flat films. Instead, the behavior is closer to two-dimensional ($\lambda = 1/\nu \approx 1$) for $L \leq 10$, and true three-dimensional scaling only emerges once the thickness exceeds about $L \approx 15$. This means that small roughness doesn't change the scaling behavior much in the thickness range where three-dimensional scaling works.

5 Conclusion

In this study, we used Monte Carlo simulations to analyze susceptibility, specific heat, and magnetization in thin Ising films for different levels of roughness and various values of N. We found that uncorrelated roughness mainly reduces the critical temperature and the height of the specific-heat peak, with noticeable effects on the susceptibility peak only for large roughness ($\Delta L \geq 1$). For films with fixed roughness, the reduced critical temperature t_L scales with L similarly to flat films, showing a crossover from two- to three-dimensional behavior at small thicknesses, while the critical exponents remain unchanged.

The dependence of t_L on L is also observed in experiments. For small roughness, which is present even under good growth conditions, the three-dimensional behavior of t_L is not affected in the thickness range where it applies ($L \geq 10$). Large variations of T_c with roughness imply that t_L should decrease smoothly with L if there is a smooth relation between L and ΔL.

Finally, simple extensions of this model, such as including correlated roughness or magnetic inhomogeneities, may help explain additional features observed in real experimental systems.

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